

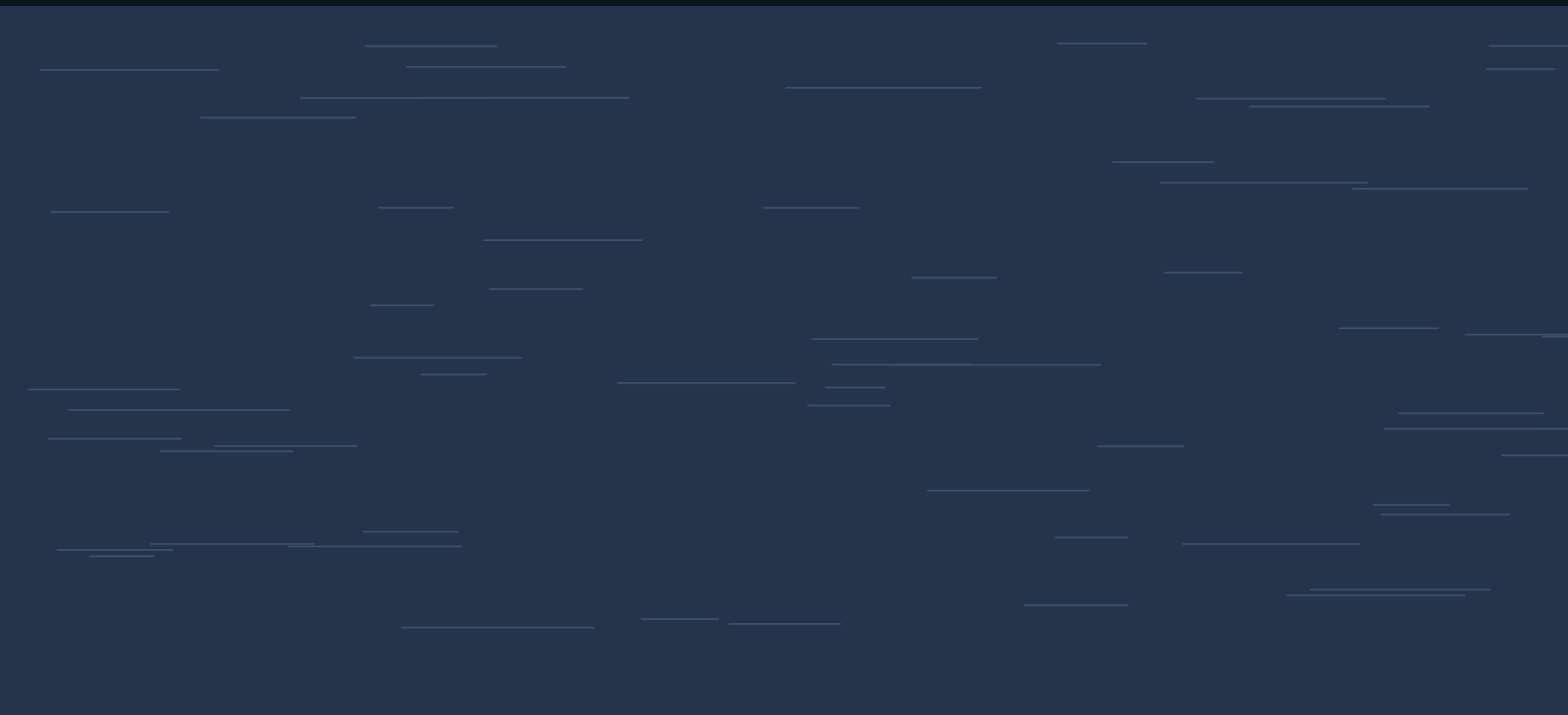


POWER LAB ESCAPE

A 3D escape room about island grids, hydrogen,
hacking and energy trading

Student manual · read before you play · no spoilers inside

Play in the browser: lnes-rh.github.io/escaperoom



1 • Your mission

Friday, 18:40. You came to the Solar Test Laboratory in Cologne to certify its island microgrid. Then a storm knocked out the public grid. The building is dark, and the door of the test booth — a motor-driven, fail-secure sliding door — will not move without power. The emergency lights run on a battery that lasts about an hour. The head of the lab, Marco Volta, has already left for the weekend.

Your job: bring the lab's own power system to life, get into the control room, and get the building back on the grid. The game is played in four stages:

Stage	Where	What you have to do
1 • The booth	Test booth	Start the lab microgrid (PV, battery, hydrogen, fuel cell) so that every phase can carry the door motor.
2 • The control PC	Control room	Find clues in the 3D world to log in to Marco's computer and become root.
3 • gridctl	Control PC	Plan 24 hours of trading for the lab's energy system to earn a reconnection permit from the grid operator.
4 • Synchronise	Tie panel Q0	Connect the lab to the public grid without a bang. Then walk out.

How to play as a team

- Play in teams of **2–3**: one person drives, the others read panels, take notes and check this manual. Swap the driver after each stage.
- Allow **45–75 minutes**. The 60-minute timer is part of the story; you can keep playing in overtime.
- **Everything you need is in the game.** This manual explains the physics and the tools, not the answers.
- Use the notes page at the end of this booklet: many clues are needed again much later.

2 • Controls

Action	Keyboard + mouse	Phone / tablet
Move • run	W A S D • hold Shift	Left thumb joystick • RUN button
Look around	Mouse (click into the game first)	Drag with the right thumb
Use / open an object	E or left click	Tap the object or USE
Hint	H	Hint button
Journal (all clues found so far)	J	Journal button
Menu, settings, pause	Esc	Menu button

The **crosshair** in the middle of the screen turns active and an object gets an amber outline when you can use it. Objects open a **panel** on the right with switches, readings or text. The game **saves automatically**; choose Continue on the title screen to resume.

3 · What you see on screen

Element	What it tells you
Timer (top left)	Time left on the emergency light, including time penalties for hints. Your rank at the end depends on it.
Objective (bottom left)	The current goal — never the solution.
Main bus box	Appears once you work on the lab devices: the headroom (spare power) on each of the three phases L1, L2, L3.
Inventory chips	Items you carry. Items are used from the panel of the object they belong to.
Journal	Every clue you have read or heard is copied here automatically, with the time you found it.

Hints

Press **H** when you are stuck. Hints always refer to your current problem and come in three steps:

Step	Cost	What you get
Nudge	free	A question that points you in the right direction.
Pointer	+1:00	Where to look and what to combine.
Solution	+3:00	The exact answer.

In the trading game an **advisor** is available for single hours; each use adds 0:15.

Settings (Esc → Settings)

Look sensitivity, invert Y, field of view, volume, UI size, brightness, **reduced motion**, **reduce flashing** (softer lightning — recommended for players sensitive to flashing light), a **colour-blind-safe** palette for the phase colours, and render quality. If the game runs slowly, choose Low quality.

Escape-room rules of thumb

- Read everything: signs, posters, labels, sticky notes, e-mails, files. Listen to the voice recorders.
- Numbers and words you find usually have exactly one purpose. Write them down with where you found them.
- If something looks broken or missing, someone probably took it away — and hid it.
- Not every object is a clue. Some just tell you about the people who work here (and their cat).
- Wrong attempts at some locks and boards cost time. Think first, then try.

4 · Science for stage 1: an island microgrid

With the public grid gone, the lab must run as an **island**. Nothing outside supplies or absorbs power, so at every instant **generation must equal consumption**. Sources only deliver what the loads actually draw.

Three phases, three islands

The lab's main bus has three phases, **L1**, **L2** and **L3**. Each of the three inverters is **single-phase** and **grid-forming**: it creates the 230 V / 50 Hz of exactly one phase you assign to it. Power cannot jump from one phase to another — a load on L2 can only be supplied by a source on L2.

Headroom = what the sources on a phase could still deliver minus what that phase already consumes. If the demand on a phase is higher than its sources can deliver, the inverter reaches its current limit, the voltage collapses and an **under-voltage relay trips** that phase. Reset it in the main bus panel once the overload is gone.

The lab's devices

Device	What it does	Key numbers
Sun simulator + PV test rig	Lamps light the III-V solar modules; the PV produces DC power.	5.4 kW DC at 3 suns. Lamp → light → PV → AC is only about 9 % efficient.
INV-1 (hybrid PV inverter)	Turns PV power into AC on one phase.	≈ 5.2 kW AC (η 97 %)
Battery + INV-2	Stores energy. Mode CHARGE (load) or DISCHARGE (source).	10 kWh, charge 2.0 kW, discharge 3.5 kW, 95 % efficient each way
PEM electrolyzer	Uses electricity to split water: $\text{H}_2\text{O} \rightarrow \text{H}_2 + \frac{1}{2} \text{O}_2$.	3 kW, 55 kWh per kg of H_2
H_2 buffer tank	Stores the hydrogen. The valve needs a handwheel.	30 bar, 0.22 kg H_2
PEM fuel cell + INV-3	Turns hydrogen back into electricity.	3.4 kW, 18 kWh of electricity per kg; needs a few seconds to warm up
Every energised phase	Lights and controls.	0.2 kW auxiliary load per phase

Merit order and storage

On each phase the sources are used in a fixed order: **PV first, then the battery, then the fuel cell**. Storage drains only by the power it actually delivers. The game runs on a **time-lapse**: 10 real seconds are one lab hour, so filling the battery or the tank takes a little real time.

Think about it

Power → hydrogen → power returns only about one third of the electricity (55 kWh in per kg, 18 kWh out per kg). Why would anyone still store energy as hydrogen?

The door

The booth door is driven by a **3-phase induction motor**. Starting it loads **every phase at the same time** with about 3 kVA for 5 seconds (a large, mostly reactive inrush current). If one phase is dead or too weak, the motor hums and stalls — this is called single-phasing. The door controller shows the headroom of each phase.

5 · Puzzle toolbox

Some locks and clues need a bit of general knowledge. The **resistor colour code** is needed in the booth (stage 1); the terminal, binary numbers, logic gates and the Caesar cipher in the control room (stage 2).

The control PC (stage 2)

The control PC is a Linux-like terminal. Type `help` to see the commands. Useful general knowledge:

Command	Meaning
<code>ls · ls -a</code>	List files · also show hidden files (their names start with a dot)
<code>cat file</code>	Show the contents of a file
<code>cd folder</code>	Change directory
<code>su user</code>	Switch user (asks for a password)
<code>mail</code>	Read the current user's e-mail

Resistor colour code (stage 1)

A 4-band resistor is read from the end its bands are closest to. **Bands 1 and 2 are digits, band 3 is a multiplier** (the number of zeros), band 4 is the tolerance.

Colour		Digit	Multiplier
black		0	×1
brown		1	×10
red		2	×100
orange		3	×1 000
yellow		4	×10 000

Colour		Digit	Multiplier
green		5	×100 000
blue		6	×1 M
violet		7	×10 M
grey		8	×100 M
white		9	×1 G

Example (not a code from the game): brown · black · orange · gold = 1, 0, ×1 000 → 10 000 Ω, ±5 %.

Binary numbers (stage 2)

A row of LEDs can be read as a binary number: lit = 1, dark = 0. The leftmost LED is the most significant bit (MSB). With 8 LEDs the place values are **128 · 64 · 32 · 16 · 8 · 4 · 2 · 1**. Example: 0000 0101 = 4 + 1 = 5.

Logic gates (stage 2)

A	B	AND	OR	XOR	NAND	NOR
0	0	0	0	0	1	1
0	1	0	1	1	1	0
1	0	0	1	1	1	0
1	1	1	1	0	0	0

NOT inverts its input. A small circle (bubble) on a gate input or output also means "inverted". Work backwards from the output: what must each gate deliver so that the final output is 1?

Caesar cipher (stage 2)

Each letter is shifted by a fixed number of places in the alphabet (with a shift of 3: A → D, B → E, ... X → A). To decode, shift back by the same number. The shift itself is hidden somewhere in the lab.

6 · Science for stage 3: trading energy (gridctl)

To get a **reconnection permit**, the lab must show the grid operator a sensible 24-hour dispatch plan for its large system (600 kWp PV, 1 MWh battery, 200 kW electrolyzer, 150 kW fuel cell, 300 kW grid connection). You set the devices hour by hour and press **Run hour**.

Term	Meaning
Day-ahead price	The forecast price for each hour, known in advance (dashed line).
Intraday price	The real price, revealed when the hour runs. If it is cloudier than forecast, there is less solar power in the region and the price goes up.
Export / import	Exported energy is paid at the price. Imports cost the price plus 40 €/MWh grid fees.
Negative prices	On sunny days there can be too much power: exporting then costs money. Curtail PV, charge, or run the electrolyzer.
Break-even prices	Shown in the forecast box: below the electrolyzer break-even, making hydrogen is worth more than selling power; above the fuel-cell break-even, burning hydrogen pays.
Green vs. grey H ₂	Hydrogen only counts as renewable (and fetches the day's H ₂ price) if it comes from your own PV surplus or from hours with a day-ahead price ≤ 20 €/MWh. Otherwise it sells as grey hydrogen at 2 €/kg.
Score	Your result is compared with a "do nothing" plan and with a perfect-foresight benchmark. You need ≥ 55 % of the benchmark's extra profit.

A simple strategy that usually passes

- Buy low, sell high: **charge the battery when power is cheap** (midday, negative prices) and **discharge at the evening peak**.
- Run the electrolyzer on PV surplus or in very cheap hours — not on expensive night-time grid power.
- Run the fuel cell only when the price is above its break-even.
- At night, when nothing is cheap or expensive, leave everything at zero and fast-forward with the "3 h" button.
- Failed? Review the day shows the benchmark hour by hour. Replay keeps the forecast, but reality comes out differently — so learn the rule, not the numbers.

Think about it

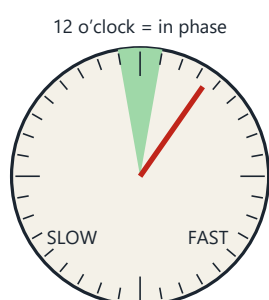
Why does a storage operator earn money exactly when the price difference between hours is large? What does this mean for a power system with a lot of solar energy?

7 · Science for stage 4: synchronising with the grid

You may only connect two AC systems when they are **in step**. Closing a breaker between two voltages that are out of phase drives huge currents — it trips immediately and, with real machines, can damage them. Before closing the tie breaker **Q0**, four conditions must hold:

#	Condition	Why
1	The same voltage (within about 2 %)	Otherwise reactive current rushes from one side to the other.
2	Almost the same frequency — the island a hair faster	If the island were slower, the grid would push power into it (reverse power) the moment the breaker closes.
3	The same phase rotation (L1 → L2 → L3)	If two phases are swapped, two poles of the breaker close onto the full line voltage.
4	The same phase angle	Close when the synchroscope needle is at 12 o'clock.

Synchroscope and dark lamps



Dark together: same rotation



Chasing each other: rotation wrong

The **synchroscope** needle turns at the difference frequency: clockwise (**FAST**) when the island is faster, anticlockwise when slower, and it stands still when both are equal. The three **lamps** are connected across the open breaker poles. When rotation and phase match, all three go dark **together**. If they light up one after another (they "chase" each other), the phase rotation is wrong. Filament lamps look dark over a wide angle — so use the lamps to check the rotation, and the synchroscope to choose the moment.

Think about it

You can only synchronise to a grid that is already live. What does that tell you about the city outside the window when you reach this stage?

Glossary

Term	Short explanation
AC / DC	Alternating current (50 cycles per second here) / direct current (from PV, batteries, fuel cells).
Phase (L1, L2, L3)	One of the three AC conductors; their voltages are shifted by 120°.
Inverter	Power electronics that turns DC into AC. Grid-forming inverters set voltage and frequency themselves.
kW / kWh	Power (how fast energy flows) / energy (power × time).
kVA	Apparent power: includes the reactive current that does no work but still loads cables and inverters.
Electrolyzer / fuel cell	Electricity → hydrogen / hydrogen → electricity.
Curtailement	Deliberately producing less PV power than possible.
Breaker / relay	A switch that can interrupt a current / a device that opens it automatically when something is wrong.

8 · Team notes

Write down every number, word or observation with the place you found it. Tick it off when you have used it.

What we found	Where	Used for	done

After the game

Question	Your answer
Why could the door not open as long as one phase had too little headroom?	
Which part of the energy chain lost the most energy, and why?	
What would you change in your trading plan if the next day were cloudy?	
Our time · hints used · rank:	